

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

Original ESAT CAMP practice material. Not an official UAT-UK or Pearson paper.

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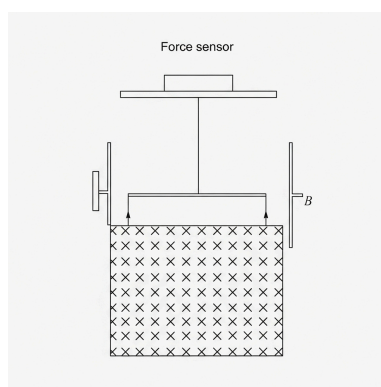
Physics

- 1 A point on a guitar string oscillates in simple harmonic motion with a frequency of 50 Hz and an amplitude of 4.0 mm.

What is the total distance travelled by this point during a time interval of 3.0 s?

- A 0.0 m
- B 0.6 m
- C 1.2 m
- D 2.4 m
- E 3.8 m

- 2 A straight, horizontal wire of length 20 cm is suspended from a sensitive force sensor. The wire is in a uniform magnetic field of strength 0.50 T, directed perpendicular to the wire. A total charge of 12 C passes through the wire over a time of 2.0 s. What is the magnitude of the change in the reading on the force sensor?



concept image

- A 0.20 N
- B 0.60 N
- C 1.2 N
- D 2.4 N
- E 3.0 N
- F 60 N

- 3 Two objects, X and Y, are in motion. Object X has 3 times the momentum of object Y and 4 times the kinetic energy of object Y.

What is the ratio of their masses, $\frac{m_X}{m_Y}$?

- A $\frac{3}{16}$
- B $\frac{3}{4}$
- C $\frac{4}{9}$
- D $\frac{9}{4}$
- E 12
- F 36

- 4 The intensity I of a physical field varies with distance r from a point source according to a power law relationship $I = kr^n$, where k and n are constants. Experimental measurements yield the following data:

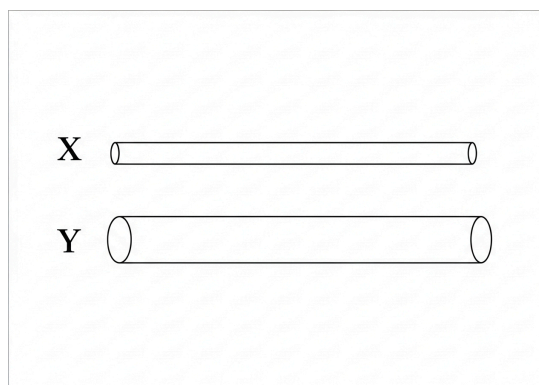
$$\text{At } r = 2.0 \text{ m, } I = 72 \text{ W m}^{-2}$$

$$\text{At } r = 4.0 \text{ m, } I = 18 \text{ W m}^{-2}$$

What is the intensity I at a distance of $r = 3.0 \text{ m}$?

- A 24 W m^{-2}
- B 32 W m^{-2}
- C 36 W m^{-2}
- D 45 W m^{-2}
- E 48 W m^{-2}

- 5 A mechanical system with natural frequency f_0 is driven by a periodic force of constant amplitude and variable frequency f . The system is lightly damped, resulting in a sharp resonance peak in the amplitude A at $f \approx f_0$. Which of the following values of the driving frequency f corresponds to the condition where the rate of change of amplitude with respect to frequency, $\frac{dA}{df}$, is **positive** and has the **largest magnitude**?
- A $0.1f_0$
 - B $0.9f_0$
 - C $1.0f_0$
 - D $1.1f_0$
 - E $2.0f_0$
- 6 Wire X has length L and diameter d . Its electrical resistance is R . Wire Y is made of the same material. It has length $2L$ and diameter $4d$. What is the electrical resistance of wire Y?



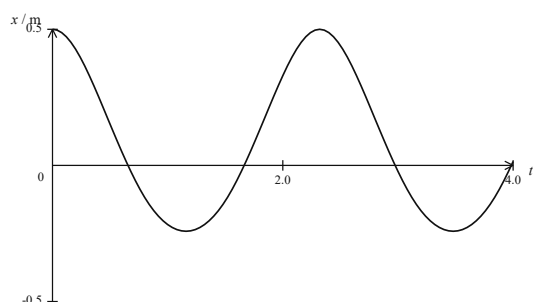
concept image

- A $R/8$
- B $R/4$
- C $R/2$
- D R
- E $2R$
- F $8R$

- 7 A circuit is constructed with five identical resistors, each of resistance R , connected to a DC supply of voltage V . The circuit consists of two parallel branches, each containing two resistors in series. A fifth identical resistor connects the midpoint of the first branch to the midpoint of the second branch. The total power dissipated by the circuit is P . If the connecting resistor between the two branches is removed, which expression gives the new total power dissipated?

- A $\frac{1}{2}P$
- B $\frac{4}{5}P$
- C P
- D $\frac{5}{4}P$
- E $2P$

- 8 The graph shows the displacement of a particle undergoing simple harmonic motion as a function of time.



What is the average speed of the particle during the first 4.0 s of its motion?

- A 0 m s^{-1}
- B 0.25 m s^{-1}
- C 0.50 m s^{-1}
- D 1.0 m s^{-1}
- E $\frac{\pi}{2} \text{ m s}^{-1}$
- F 2.0 m s^{-1}

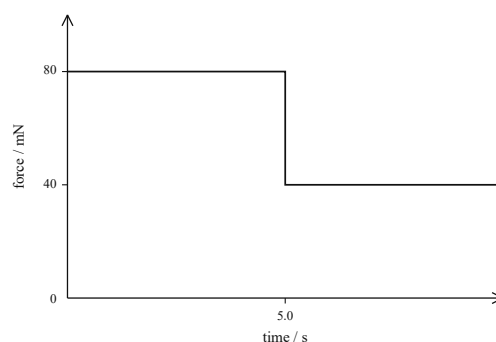
- 9 A block of weight W rests on a rough plane inclined at 45° to the horizontal. The coefficient of static friction between the block and the plane is 0.5.

A horizontal force P is applied to the block.

Which inequality gives the range of values of P for which the block remains stationary?

- A $\frac{1}{2}W \leq P \leq \frac{3}{2}W$
 B $\frac{\sqrt{2}}{4}W \leq P \leq \frac{3\sqrt{2}}{4}W$
 C $\frac{1}{3}W \leq P \leq \frac{3}{2}W$
 D $\frac{1}{3}W \leq P \leq 3W$

- 10 A small charged sphere is suspended by a thread from a force sensor. The sphere is in a uniform vertical electric field of strength $5.0 \times 10^3 \text{ N C}^{-1}$. The graph shows the reading on the force sensor over time. Initially the reading is constant at 80 mN. At time $t = 5.0 \text{ s}$, the direction of the electric field is reversed, and the reading changes to a new constant value of 40 mN.



What is the magnitude of the charge on the sphere?

- A $2.0 \mu\text{C}$
 B $4.0 \mu\text{C}$
 C $8.0 \mu\text{C}$
 D $12 \mu\text{C}$
 E $16 \mu\text{C}$
 F $4000 \mu\text{C}$

- 11** A string fixed at both ends is observed to resonate at frequencies of 150 Hz and 250 Hz. These are not consecutive resonant frequencies.

What is the fundamental frequency of the string?

- A** 25 Hz
- B** 50 Hz
- C** 100 Hz
- D** 150 Hz
- E** 200 Hz

- 12** An electric kettle has a power input of 2.5 kW. When used to boil water, it produces steam at a steady rate of 1.0 g s^{-1} .

What is the rate of energy loss from the kettle to the surroundings?

(Specific latent heat of vaporisation of water = $2.0 \times 10^6 \text{ J kg}^{-1}$; specific heat capacity of water = $4.2 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$)

- A** 0 W
- B** 500 W
- C** 1500 W
- D** 2000 W
- E** 2500 W
- F** 4500 W

- 13** A nucleus of Thorium-232 (${}_{90}^{232}\text{Th}$) decays through a series of emissions to become a stable nucleus of Lead-208 (${}_{82}^{208}\text{Pb}$).

The decay series consists entirely of alpha (α) particles and beta-minus (β^-) particles.

What are the total numbers of α and β^- particles emitted during this process?

- A** 4 α , 6 β^-
- B** 6 α , 4 β^-
- C** 6 α , 8 β^-
- D** 6 α , 12 β^-

- 14** A small heating element at the bottom of an open tank of liquid heats a small parcel of the liquid immediately above it.

Just before this heated parcel begins to rise, how do its mass, weight, and the upthrust acting on it compare to their values before it was heated?

- A** Mass: Unchanged, Weight: Unchanged, Upthrust: Increases
- B** Mass: Unchanged, Weight: Unchanged, Upthrust: Decreases
- C** Mass: Unchanged, Weight: Decreases, Upthrust: Unchanged
- D** Mass: Decreases, Weight: Decreases, Upthrust: Unchanged
- E** Mass: Decreases, Weight: Decreases, Upthrust: Increases
- F** Mass: Unchanged, Weight: Decreases, Upthrust: Increases

- 15** Two identical copper spheres, **X** and **Y**, are heated to 500 K.

Sphere **X** has a matte black surface.

Sphere **Y** has a polished silver surface.

The initial rate of energy loss, R , is determined for each sphere in a **vacuum** and in an atmosphere of **argon** gas at 298 K.

In the vacuum, it is observed that $R_X > R_Y$.

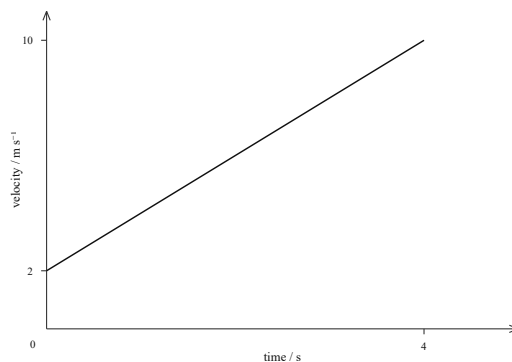
Assume that the rate of heat transfer due to convection depends only on the dimensions of the sphere and the temperature difference.

Which statement correctly describes the rates in the argon atmosphere compared to the vacuum?

- A** The rates R_X and R_Y increase by the same amount, and the ratio $\frac{R_X}{R_Y}$ decreases.
- B** The rates R_X and R_Y increase by the same amount, and the ratio $\frac{R_X}{R_Y}$ stays constant.
- C** The rate R_X increases more than R_Y because black surfaces are better thermal conductors, and the ratio $\frac{R_X}{R_Y}$ increases.
- D** The rate R_Y increases more than R_X because the polished surface reflects gas molecules, and the ratio $\frac{R_X}{R_Y}$ decreases.

- 16** A continuous wave travels from medium X to medium Y. The wave speed in medium Y is $\frac{2}{3}$ of the wave speed in medium X.
What is the fractional change in the wavelength of the wave?
- A** $-\frac{1}{3}$
B 0
C $\frac{1}{3}$
D $\frac{1}{2}$
E $\frac{2}{3}$
- 17** The radioactive nuclide ${}_{92}^{235}\text{U}$ decays into the stable nuclide ${}_{82}^{207}\text{Pb}$ via a series of α (alpha) and β^- (beta-minus) emissions.
Which option gives the total number of α and β^- particles emitted during this decay chain?
- A** 5 α , 0 β^-
B 7 α , 4 β^-
C 7 α , 10 β^-
D 8 α , 6 β^-
- 18** An electric heater of constant power is used to heat a metal block. The block is initially at room temperature, 20°C .
After 10 minutes, the temperature of the block is 50°C .
The rate of thermal energy loss to the surroundings is proportional to the temperature difference between the block and the surroundings.
Which of the following describes the temperature θ of the block after a total of 20 minutes?
- A** $\theta = 50^\circ\text{C}$
B $50^\circ\text{C} < \theta < 80^\circ\text{C}$
C $\theta = 80^\circ\text{C}$
D $80^\circ\text{C} < \theta < 100^\circ\text{C}$
E $\theta = 100^\circ\text{C}$

- 19** An object of mass 2.0 kg moves in a straight line. The velocity-time graph for its motion is shown.



What is the magnitude of the constant net force acting on the object?

- A** 1.0 N
 - B** 2.0 N
 - C** 4.0 N
 - D** 5.0 N
 - E** 12 N
 - F** 20 N
- 20** A cylindrical metal wire has an electrical resistance R . The wire is uniformly stretched until its length is 3 times its original length.

Assume that the volume of the wire remains constant and the resistivity of the metal does not change.

What is the resistance of the stretched wire?

- A** $\frac{R}{9}$
- B** $\frac{R}{3}$
- C** $3R$
- D** $9R$
- E** $27R$

- 21** A uniform rod of mass m and length $2L$ rests with one end on a rough horizontal floor and the other end against a smooth vertical wall. The rod is inclined at 45° to the horizontal. A particle of mass m is attached to the upper end of the rod.

Which of the following is the minimum coefficient of static friction between the rod and the floor required to maintain equilibrium?

- A** 0.25
- B** 0.5
- C** 0.75
- D** 1.0
- E** 1.5

- 22** Twelve identical resistors are connected to form the edges of a cube. A current enters the network at one vertex and leaves at the diagonally opposite vertex.

Let P_{in} be the power dissipated in a single resistor connected directly to the input vertex.

Let P_{mid} be the power dissipated in a single resistor that is connected to neither the input nor the output vertex.

What is the value of the ratio $\frac{P_{\text{in}}}{P_{\text{mid}}}$?

- A** 1
- B** 2
- C** 4
- D** 9
- E** 16

- 23** A solid composite object has total volume $4.0 \times 10^{-3} \text{ m}^3$ and is made from materials X and Y. Their densities are 2000 kg m^{-3} and 800 kg m^{-3} respectively.

It is fully submerged and neutrally buoyant in a fluid of density 1100 kg m^{-3} . For this question, neutral buoyancy means that the object's total mass equals the mass of the displaced fluid:

$$m_{\text{total}} = \rho_{\text{fluid}} V_{\text{total}}.$$

What is the mass of material X?

- A** 1.1 kg
- B** 2.0 kg
- C** 2.4 kg
- D** 4.0 kg
- E** 4.4 kg
- F** 6.0 kg

- 24** A small object attached to a vertical spring oscillates with simple harmonic motion. The amplitude of the oscillation is 5.0 cm and the frequency is 2.0 Hz.
What is the total distance travelled by the object in a time interval of 1.0 minute?
- A** 0 m
B 0.40 m
C 6.0 m
D 12 m
E 24 m
- 25** A fixed mass of an ideal gas initially has pressure P and volume V . The gas is compressed at constant temperature to a final volume of $V/3$.
Which expression gives the final pressure of the gas?
- A** $\frac{P}{9}$
B $\frac{P}{3}$
C P
D $3P$
E $9P$
- 26** A block of mass 10 kg is pulled up a rough plane inclined at an angle θ to the horizontal, where $\tan \theta = 0.75$. The rope pulling the block is parallel to the plane, and the block moves at a constant speed.
The coefficient of kinetic friction between the block and the plane is 0.25. The acceleration of free fall is $g = 10 \text{ m s}^{-2}$.
What is the tension in the rope?
- A** 40 N
B 60 N
C 80 N
D 85 N
E 95 N

- 27** A DC power supply of voltage V and negligible internal resistance is connected in series with two resistors of resistances R and $2R$.

Which expression gives the current flowing through the resistor of resistance R ?

- A** $\frac{V}{3R}$
- B** $\frac{V}{2R}$
- C** $\frac{2V}{3R}$
- D** $\frac{V}{R}$
- E** $\frac{3V}{2R}$